

## CAPCITY - Capital City

#scc (/problems/tag/scc)

There are  $N$  cities in Flatland connected with  $M$  unidirectional roads. The cities are numbered from 1 to  $N$ . The Flat Circle of Flatland (FCF) wants to set up a new capital city for his kingdom. For security reasons, *the capital must be reachable from all other cities* of Flatland. FCF needs the list of *all* candidate cities. You are the chief programmer at FACM (Flat Association for Computing Machinery) responsible for providing the list to FCF as soon as possible.

### Input

The first line of the input file contains two integers:  $1 \leq N \leq 100,000$  and  $1 \leq M \leq 200,000$ . Each of the following  $M$  lines contains two integers  $1 \leq A, B \leq N$  denoting a road from  $A$  to  $B$ .

### Output

The output file contains an integer denoting the number of candidate cities followed by the list of candidate cities in increasing order.

### Example

**Input:**

```
4 4
1 2
3 2
4 3
2 1
```

**Output:**

```
2
1 2
```

 **Submit solution!** (/submit/CAPCITY/)

|                                    |                                 |   |                                 |
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sbarreto (/users/sbarreto): 2025-04-04 05:37:15

If there where no weak test cases the solution is find the condensed graph. Revert the edges and find topSort. Lets call S the first node in that topSort, if we can reach all nodes from S mean every node can reach S, so that would be our solution. Else, there is no solution

**Last edit: 2025-04-04 05:38:03**



mr\_gentle (/users/mr\_gentle): 2024-08-08 23:35:17

poor test case,  
there is only one ssc, and we have to print that only, in increasing order



romicro (/users/romicro): 2023-10-23 18:00:16

dont forget: you should print candidate cities in increasing order.



yugalpradhan (/users/yugalpradhan): 2023-03-31 22:09:01

In question there should be given that at least one city is reachable from all other cities, or they need to change the test cases, include test cases where answer is 0.



ma7moud\_7amdy (/users/ma7moud\_7amdy): 2022-06-21 21:15:41

There are missing test cases for this problem

8 9  
1 2  
2 3  
3 4  
4 5  
5 2  
1 6  
6 7  
7 8  
8 6

test like this , there are two SCC and they can't reach each other, I have submitted a solution its result is the nodes on the two SCC and this doesn't seem right because the answer should be 0

here's the code that gets wrong result <snip>

[Simes]: Please use the forum for this.

**Last edit: 2022-06-22 21:39:23**



hoangptch (/users/hoangptch): 2021-09-30 08:52:45

4 3 4 1 1 1 3 2 1 2 2 2 3 2 4 3 2 3 3

8 5 11 1 3 1 5 2 2 2 3 2 5 2 8 3 4 3 5 3 6 3 7 4 3 4 6 5 1 5 6 6 1 6 2 6 4 6 5 7 2 7 5 7 7  
7 8

Output

#1 3

#2 2

**Last edit: 2022-06-22 08:06:26**



below\_average (/users/below\_average): 2021-07-10 13:19:58

nice problem



lakshya1st (/users/lakshya1st): 2021-05-17 15:24:30

Same as TOUR only difference is find outdegree instead of indegree in this problem...!! SCC :)



saurabh\_kl (/users/saurabh\_kl): 2021-02-26 18:41:17

A nice problem



ya\_al313 (/users/ya\_al313): 2021-01-14 19:09:08

dont forget: you should print candidate cities in increasing order.

 [Submit solution! \(/submit/CAPCITY/\)](/submit/CAPCITY/)

Added by: Narek Saribekyan  
(/users/narek)  
Date: 2010-06-20  
Time limit: 1s-4s  
Source limit: 50000B  
Memory limit: 1536MB  
Cluster: Cube (Intel G860) (/clusters/)  
Languages: All except: NODEJS OBJC  
PERL6 SQLITE VB.NET  
Resource: Armenian TST 2010, Round 2

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